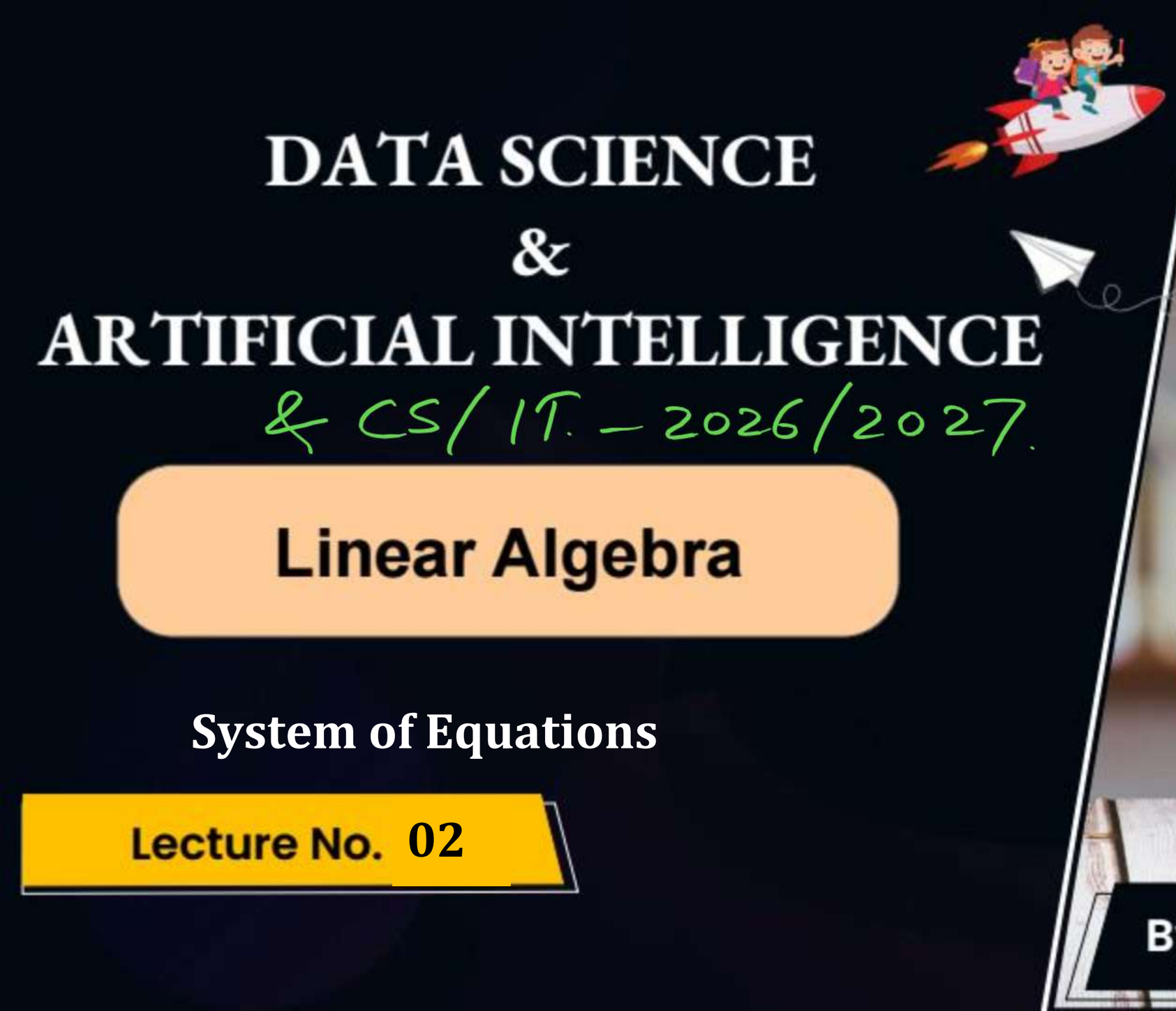
DATA SCIENCE & ARTIFICIAL INTELLIGENCE

& CS/IT. – 2026/2027.

Linear Algebra

System of Equations

Lecture No. 02



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Recap of previous lecture



Topic Non Homogeneous S.O.Eq

Topics to be Covered



Topic

1. Remaining Part of Non Homog. System of Equ's
2. Homogeneous System

METHODS of SOLVING Non Homog. System

$(A_{m\times n}X_{n\times 1} = B_{m\times 1})$

① RANK Method (always applicable)

(i.e. for $m > n$, $m = n$, $m < n$)

(i) if $\boxed{\rho(A) = \rho(A:B)} =$ No. of Variables $\Rightarrow$ unique Sol^n

(ii) if $\boxed{\rho(A) = \rho(A:B)} <$ " " $\Rightarrow$ Infinite Sol^n

(iii) if $\rho(A) \neq \rho(A:B) \Rightarrow$ No Solution.

Matrix Method

(applicable only when $m = n$)

(i) if $|A| \neq 0 \Rightarrow$ unique Sol^n exist

(ii) if $|A| = 0$ & $(\text{adj}\,A)B = 0 \Rightarrow$ (∞ Sol^n.)

(iii) if $|A| = 0$ & $(\text{adj}\,A)B \neq 0 \Rightarrow$ (No Sol^n)

$AX = B \Rightarrow X = A^{-1}B = \dfrac{(\text{adj}\,A)B}{|A|}$

(*) $\boxed{\rho(A) \neq \rho(A:B)}$ $\Rightarrow$ Rank of Coeff Mat can never exceed Rank of Augmented Mat.

$$\begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & 1 & 2 & : & 4 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

$\rho(A) = 2 = \rho(A:B)$

(∞ Sol^n)

$$\begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & 1 & 2 & : & 4 \\ 0 & 0 & 5 & : & 0 \end{bmatrix}$$

$\rho(A) = 3 = \rho(A:B)$ (Unique Sol^n)

$$\begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & 1 & 2 & : & 4 \\ 0 & 0 & 0 & : & 5 \end{bmatrix}$$

$\rho(A) = 2$, $\rho(A:B) = 3$

(No Sol^n)

$$\begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & 1 & 2 & : & 4 \\ 0 & 0 & 5 & : & 14 \end{bmatrix}$$

(Unique Sol^n)

$\rho(A) = 3 = \rho(A:B)$

(Ex) Underdetermined System Can not Have Unique Solution (Learn)

eg underdetermined System if Consistent always consist ∞ Sol.

& " " " if Inconsistent contains No Sol.

Take Care :- overdetermined System Can have all types of Solutions.

ie we may get unique Sol, ∞ Sol. or No Solution

& we have to follow only Rank Method

(P8) Consider the system $\begin{bmatrix} 2 & 1 & -4 \\ 4 & 3 & -12 \\ 1 & 2 & -8 \end{bmatrix}\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} \alpha \\ 5 \\ 7 \end{bmatrix}$. Notice that 2nd & 3rd column

PYQ

of Coeff Mat are L.D then for how many values of α, does the above system has ∞ solutions;

$A \cdot X = B$

(a) 1

(b) 3

(c) ∞

(d) $1/5$

$$[A:B] = \begin{bmatrix} 2 & 1 & -4 & : & \alpha \\ 4 & 3 & -12 & : & 5 \\ 1 & 2 & -8 & : & 7 \end{bmatrix}_{3\times 4} \xrightarrow{R_1 \leftrightarrow R_3} \begin{bmatrix} 1 & 2 & -8 & : & 7 \\ 4 & 3 & -12 & : & 5 \\ 2 & 1 & -4 & : & \alpha \end{bmatrix}$$

$$\xrightarrow[R_3 \to R_3 - 2R_1]{R_2 \to R_2 - 4R_1} \cdots\cdots\cdots \begin{bmatrix} 1 & 2 & -8 & : & 7 \\ 0 & a_{22} & a_{23} & : & a_{24} \\ 0 & 0 & 0 & : & \frac{5\alpha - 1}{2} \end{bmatrix}_{3\times 4}$$

$\therefore \rho(A) = 2$ then for ∞ soln, $\rho(A:B) = 2 \Rightarrow \frac{5\alpha - 1}{2} = 0 \Rightarrow \alpha = \frac{1}{5}$

is only one value of α exist for ∞ soln..

Q: if $A_{n\times n}$ s.t $A^2 = I$ then $AX = B$ has _____ solutions

(a) unique sol
(b) ∞ sol
(c) No sol
(d) More than one but finite No. of sol

∵ No. of eqn = No of Variables i.e system is equally determined
so we can apply Matrix Method also.

Given $A^2 = I$
$\Rightarrow A^{-1} = A$
i.e A^{-1} exist
or $|A| \neq 0$
⇓
By Matrix Method unique sol. exist.

M-II $A^2 = I \Rightarrow |A| = 1$ or -1 i.e $|A| \neq 0$
so By Mat Method, unique sol exist

M-III By Common Sense :-

Given, $AX = B$
$A(AX) = AB$
$A^2X = AB$
$X = AB = \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix}_{n\times 1}$ = unique vector exist.

Unique sol exist

Q. If $A_{3\times4}$ Mat s.t $AX=B$ is an Inconsistent system then Highest possible Rank of A will be?

(a) 1
(b) 2
(c) 3
(d) C.N.D

Given $A_{3\times4} \Rightarrow [A:B]_{3\times5}$

& For Inconsistency,

$\rho(A) \neq \rho(A:B)$

$\Rightarrow \rho(A) < \rho(A:B)$

Now $\text{Max}\,\rho(A:B) = 3$

$\Rightarrow \text{Max}\,\rho(A) = 2$

M-II For Inconsistency;

$\rho(A) < \rho(A:B)$

So Various possible Cases are;

Case I: $\rho(A)=2,\ \rho(A:B)=3$

Case II: $\rho(A)=1,\ \rho(A:B)=3$

Case III: $\rho(A)=1,\ \rho(A:B)=2$

So $\text{Max}\,\rho(A) = 2$

Q: The system,

$$x_1 - 2x_2 + 4x_3 = 5$$
$$-3x_1 + 6x_2 - 12x_3 = -15$$

has ∞ soln.

$$[A:B] = \begin{bmatrix} 1 & -2 & 4 & : & 5 \\ -3 & 6 & -12 & : & -15 \end{bmatrix}_{2\times 4}$$

$R_2 \to R_2 + 3R_1$

$$= \begin{bmatrix} 1 & -2 & 4 & : & 5 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

$\rho(A) = 1, \rho(A:B) = 1$

$\because \rho(A) = \rho(A:B) < \text{No of Variables}(3)$

$\Rightarrow \infty$ soln exist.

Q: The system,

$$x_1 - 2x_2 + 4x_3 = 5$$
$$-3x_1 + 6x_2 - 12x_3 = 7$$

has ——— soln.

$$[A:B] = \begin{bmatrix} 1 & -2 & 4 & : & 5 \\ -3 & 6 & -12 & : & 7 \end{bmatrix}_{2\times 4}$$

$R_2 \to R_2 + 3R_1$

$$[A:B] \sim \begin{bmatrix} 1 & -2 & 4 & : & 5 \\ 0 & 0 & 0 & : & 22 \end{bmatrix}$$

$\because \rho(A) = 1$ & $\rho(A:B) = 2$

$\Rightarrow$ No soln exist

Q: Solve $x-y+2z=1$, $2x+y-z=2$, $3x+z=3$, $x+2y-3z=1$

(HW)

$$[A:B]=\left[\begin{array}{ccc:c} 1 & -1 & 2 & 1 \\ 2 & 1 & -1 & 2 \\ 3 & 0 & 1 & 3 \\ 1 & 2 & -3 & 1 \end{array}\right]$$

$$\sim\left[\begin{array}{ccc:c} 1 & -1 & 2 & 1 \\ 0 & 3 & -5 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$\rho(A)=2,\ \rho(A:B)=2$

$\because \rho(A)=\rho(A:B)<$ No. of Variables $(3) \Rightarrow \infty$ sol exist

Process of finding sol:-

$$\begin{bmatrix} 1 & -1 & 2 \\ 2 & 1 & -1 \\ 3 & 0 & 1 \\ 1 & 2 & -3 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}=\begin{bmatrix} 1 \\ 2 \\ 3 \\ 1 \end{bmatrix}$$

$$\sim\begin{bmatrix} 1 & -1 & 2 \\ 0 & 3 & -5 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}=\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} (x-y+2z) \\ (3y-5z) \\ 0 \\ 0 \end{bmatrix}=\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$\Rightarrow x-y+2z=1$ & $3y-5z=0$

Let $z=k$, $3y=5z \Rightarrow y=\frac{5k}{3}$

$x-y+2z=1$

$x-\frac{5k}{3}+2k=1 \Rightarrow x=1-\frac{k}{3}$

$$X=\begin{bmatrix} x \\ y \\ z \end{bmatrix}=\begin{bmatrix} 1-\frac{k}{3} \\ \frac{5k}{3} \\ k \end{bmatrix}$$

$$=\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}+\begin{bmatrix} -k/3 \\ 5k/3 \\ k \end{bmatrix}$$

$$X = \begin{bmatrix}1\\0\\0\end{bmatrix} + \begin{bmatrix}-k/3\\5k/3\\k\end{bmatrix}$$

Let $k = 3\alpha$, $\alpha \in R$

$$X = \begin{bmatrix}1\\0\\0\end{bmatrix} + \begin{bmatrix}-\alpha\\5\alpha\\3\alpha\end{bmatrix}$$

$$X = \begin{bmatrix}1\\0\\0\end{bmatrix} + \alpha\begin{bmatrix}-1\\5\\3\end{bmatrix}$$

i.e. For $\alpha = 0$, $X_1 = \begin{bmatrix}1\\0\\0\end{bmatrix}$

For $\alpha = 1$, $X_2 = \begin{bmatrix}0\\5\\3\end{bmatrix}$

For $\alpha = -1$, $X_3 = \begin{bmatrix}2\\-5\\-3\end{bmatrix}$

For $\alpha = 2$, $X_4 = \begin{bmatrix}-1\\10\\6\end{bmatrix}$ & so on - - - - - -

A set of ∞ soln^{n} is given as $\left\{\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\5\\3\end{bmatrix}, \begin{bmatrix}2\\-5\\-3\end{bmatrix}, \begin{bmatrix}-1\\10\\6\end{bmatrix} - - - - - -\right\}$

Homogeneous System ($AX=O$)

eg $\left.\begin{aligned} 2x-3y+5z&=0\\ x+4y+2z&=0\\ 7x-2y-z&=0\\ 4x-y+4z&=0 \end{aligned}\right\}$ $X=\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}$ is the sol of this system (T)

& this sol. always exist that's why

we say that, Homog. System NEVER Inconsistent

OR

Homog System ALWAYS Consistent.

① unique sol exist ≃ Trivial sol exist ≃ Zero sol always exist.

② ∞ sol exist ≃ Non Trivial sol also exist ≃ Non Zero sol also exist.

③ No sol^{n} $\neq$ Zero sol^{n}. (in Homog system)

$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} ? \\ ? \\ ? \end{bmatrix}$ $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

④ Underdetermined Homog system always consist ∞ sol^{n}.



⑤ In Case of Homog system, $\rho(A) = \rho(A:0)$ always

So No Need to write Augmented Matrix further.

Methods of solving Homog system

$(A_{m\times n} X_{n\times 1} = O_{m\times 1})$

① RANK Method (always applicable)

$(m > n,\ m = n,\ m < n)$

(i) If $\rho(A)$ = No. of Variables $\Rightarrow$ unique sol.

(ii) If $\rho(A) <$ " " $\Rightarrow \infty$ sol.

② MATRIX Method

(applicable only when $m = n$)

(i) $|A| \neq 0 \Rightarrow$ unique sol.

(ii) $|A| = 0 \Rightarrow \infty$ sol.

Note:- ① unique sol $\simeq$ Trivial sol $\simeq$ Zero sol.

② ∞ sol $\simeq$ Non Trivial sol $\simeq$ Non Zero sol.

✱ Nullity of Homog System →

(Dimension of Null space)

Nullity = Counting of (L.I solutions) for Homog System $AX = O$

✱ RANK-Nullity Th :

$$\text{Nullity} = \text{No. of Columns} - \rho(A)$$

(Learn by Heart)

Sp Case : if we have (equally determined) Homog System i.e $A_{n\times n}$

then Nullity = No. of Zero Rows in Echelon form.

Q. Find k for which $(3k-8)x+3y+3z=0$, $3x+(3k-8)y+3z=0$, $3x+3y+(3k-8)z=0$ has Non Trivial soln's

msg

(a) 2/3
(b) 8/3
(c) 11/3
(d) 0

$$\begin{vmatrix} (3k-8) & 3 & 3 \\ 3 & (3k-8) & 3 \\ 3 & 3 & (3k-8) \end{vmatrix} = 0$$

$C_1 \to C_1+C_2+C_3$

$$\begin{vmatrix} 3k-2 & 3 & 3 \\ 3k-2 & 3k-8 & 3 \\ 3k-2 & 3 & 3k-8 \end{vmatrix} = 0$$

Has Non Trivial soln's $\Rightarrow |A| = 0$

$$(3k-2)\begin{vmatrix} 1 & 3 & 3 \\ 1 & 3k-8 & 3 \\ 1 & 3 & 3k-8 \end{vmatrix} = 0$$

R_2-R_1 & R_3-R_1

$$(3k-2)\begin{vmatrix} 1 & 3 & 3 \\ 0 & 3k-11 & 0 \\ 0 & 0 & 3k-11 \end{vmatrix} = 0$$

$(3k-2)(3k-11)^2 = 0 \Rightarrow k = \frac{2}{3}, \frac{11}{3}$

Q Consider $px+qy+rz=0$

(MSQ) $qx+ry+pz=0$

$rx+py+qz=0$

then For the existence of Non Zero (sol) choose the correct options ?

(a) $p+q+r=0$

(b) $p=q=r$

(c) $p^3+q^3+r^3-3pqr=0$

(d) None

∞ sol ⇒ $|A|=0$

$$\begin{vmatrix} p & q & r \\ q & r & p \\ r & p & q \end{vmatrix}=0$$

$C_1 \to C_1+C_2+C_3$;

$$\begin{vmatrix} p+q+r & q & r \\ p+q+r & r & p \\ p+q+r & p & q \end{vmatrix}=0$$

$$(p+q+r)\begin{vmatrix} 1 & q & r \\ 1 & r & p \\ 1 & p & q \end{vmatrix}=0$$

R_2-R_1 & R_3-R_1

$$(p+q+r)\begin{vmatrix} 1 & q & r \\ 0 & r-q & p-r \\ 0 & p-q & q-r \end{vmatrix}=0$$

$$(p+q+r)\left[-p^2-q^2-r^2+pq+qr+rp\right]=0$$

$-(p+q+r)(p^2+q^2+r^2-pq-qr-rp)=0$ i.e. (a) is Correct

$-(p^3+q^3+r^3-3pqr)=0$ i.e. (c) " "

$\rightarrow \frac{1}{2}(p+q+r)\left[2p^2+2q^2+2r^2-2pq-2qr-2rp\right]=0$

$(p+q+r)\left[(p-q)^2+(q-r)^2+(r-p)^2\right]=0$

i.e. at $p=q=r$ it is verified so (b) is also correct

PODCAST: Find the Nature of Sol of $x+y+z=0$?

Sol: $x+y+z=0$

$$\underset{1\times 3}{\begin{bmatrix}1 & 1 & 1\end{bmatrix}}\underset{3\times 1}{\begin{bmatrix}x\\y\\z\end{bmatrix}}=\underset{1\times 1}{[0]}$$

$$\boxed{A_{1\times 3}X_{3\times 1}=O_{1\times 1}}$$

i.e. $X=\begin{bmatrix}x\\y\\z\end{bmatrix}$ = Sol of above system of we will get ∞ Sol.

& these Sol. are given as,

$$X=\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix},\begin{bmatrix}-1\\0\\1\end{bmatrix},\begin{bmatrix}1\\0\\-1\end{bmatrix},\begin{bmatrix}2\\-1\\-1\end{bmatrix},\begin{bmatrix}0\\-1\\1\end{bmatrix},\begin{bmatrix}-5\\3\\2\end{bmatrix},\ \text{-----} \ \infty \text{ Sol. exist.}$$

ZERO Sol. | Non Zero Sol / ∞ Sol.

Let us have more PODCAST for Non Trivial Soln's ;

$$X = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ +1 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ -1 \end{bmatrix}, \begin{bmatrix} 3 \\ -1 \\ -2 \end{bmatrix}, \ldots\ldots\ldots\ldots \text{ soln.}$$

X_1 " X_2 " X_3 = X_4 = X_5 =

out of these ∞ soln's only X_1 & X_2 are (L.I) & Rest are LD on them.

for $X_3 = X_1 - X_2$ ie X_3 is L.D on X_1 & X_2

again, $X_4 = X_1 + X_2$ ie X_4 is LD on X_1 & X_2

" $X_5 = X_1 + 2X_2$ ie X_5 is LD on X_1 & X_2 & so on - - - -

Conclusion : the system $x+y+z=0$ has Two LI Soln.

Q: the Number of L.I Sol. of $x+y+z=0$ is/are _____

(M-I) Using Conventional App → Time Consuming

(M-II) Using Shortcut → $A=[1\ 1\ 1]_{1\times 3}$ → No. of Columns $= 3$, $\rho(A) = 1$

So Nullity = No. of Columns − $\rho(A)$ $= 3-1=2$

i.e No. of L.I Sol^{n} (for $AX=0$) = Two.

Q: If $A_{4\times 5}$ s.t Every soln of $AX=O$ ... (1) is the scalar Multiplication of $\begin{bmatrix}2\\-1\\3\\1\\0\end{bmatrix}$ then $\rho(A)=?$

(a) 5
(b) 4
(c) 1
(d) ∞

Soln of (1) is $X=\begin{bmatrix}2\\-1\\3\\1\\0\end{bmatrix} \longrightarrow \underbrace{\begin{bmatrix}4\\-2\\6\\2\\0\end{bmatrix}, \begin{bmatrix}-2\\1\\-3\\-1\\0\end{bmatrix}, \begin{bmatrix}2/3\\-1/3\\1\\1/3\\0\end{bmatrix}, \dots\dots \infty \text{ soln}}_{\text{All are LD on } X}$

ie Nullity of A = No. of LI soln = one

Hence $N(A)$ = No. of Columns $-\rho(A)$

$\rho(A) = 5-1 = 4$

THANK - YOU

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